Programming Assignment: Machine Problem 2 : Load Balancing + Auto Scaling

Graded • 4/4 points

Deadline Pass this assignment by Feb 23, 11:59 PM CST

1. Overview

This MP has two sections. One is about AWS Elastic Load Balancing (ELB), and the other is about Amazon EC2 Auto Scaling.

AWS Elastic Load Balancing (ELB) spreads user traffic across many instances of your applications. A load balancer decreases the possibility of performance issues in your applications by distributing the load. It also helps with the reliability and scalability of your product, which may be running on several virtual machines. AWS EC2 enables you to create virtual machines on AWS infrastructure that run various operating systems, including multiple flavors of Linux, Windows Server, and macOS. In the first section of this MP, we will launch load balancer to balance requests between 2 web servers hosted on Amazon EC2.

Amazon EC2 Auto Scaling is an AWS feature that automatically optimizes the number of EC2 instances availa to your workloads or app based on workloads' performance and pre-defined conditions. In the second sectic of this MP, we'll create an AWS auto scaling group to monitor the CPU usage of the EC2 instances and decide when to scale up the number of instances.

Note: To solve this problem you may want to familiarize yourself with IAM, which are covered in week 3. read

2. Requirements